

Status Update

CELLAIR Mesoscope

Parts arrived · optics redesigned · next steps

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2026-05-12

Follow-up to elevator pitch (01_elevator_pitch)

Recap — Where We Left Off

Last time: Elevator pitch presented the experimental concept (hFOB 1.19 / hMEC 1 co-culture in a perfused bioreactor) and the intended observation platform (OpenFlexure microscope on RPi 5 + Hailo-8).

Open at the time:

- ▶ Sangaboard, motors, optics kit on order
- ▶ Tube lens (125–150 mm) received
- ▶ Custom optics module *in development*
- ▶ Conventional microscope used for first images

Today's update:

- ▶ Hardware shipment received
- ▶ Custom optics STLs generated
- ▶ Camera-side interface had to be redesigned
- ▶ Path to first OpenFlexure image is now mechanical

Parts Arrived

Hardware procurement: complete for the optics + stage subsystem.

Part	Source	Status
Sangaboard v5	Taulab	✓ Received
OFMv7 Illumination Kit	Taulab	✓ Received
OFMv7 Bag of Bits	Taulab	✓ Received
3× 28BYJ-48 motors	Taulab	✓ Received
Tube lens (150 mm)	Distributor	✓ Received
Tube lens (125 mm)	Distributor	✓ Received
HQ Camera (IMX477)	–	✓ Pre-existing
RPi 5 + Hailo-8	–	✓ Operational
≤10x objective	–	✓ Pre-existing

All optical parts on hand.

Nothing left to wait for.

Path to first light is now purely mechanical
+ a print run.

Ready to print today:

- ▶ Main body, gears, condenser
- ▶ Custom optics module (150 mm)
- ▶ Tools (lens, gear)

First Obstacle — Camera-Side Interface

The upstream HQ Camera mount is wrong for our build.

OpenFlexure's `picamera_hq` mount assumes the user has *stripped the C-mount lens off* and screws the bare PCB onto four M2 posts — like the Pi Camera v2 design. We use the HQ Camera *intact*:

- ▶ Mechanical: the C-mount housing is a precise machined optical reference.
- ▶ Practical: removing the lens voids the camera and risks the IR-cut filter and sensor stack.

Symptoms encountered:

- ▶ `picamera_hq_cover.stl` dimensions don't match the camera body.
- ▶ Optics-module bottom face designed for 38×38 mm bare PCB.
- ▶ Sensor-to-tube-lens distance off by ~ 14 mm.

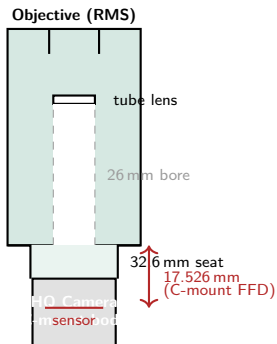
Root cause:

- ▶ Upstream: `mount_height = 4.5`, `sensor_height = 1`.
- ▶ Sensor lands 3.5 mm below mount face.
- ▶ C-mount FFD is **17.526 mm** — sensor must be 17.526 mm below.

The Fix — C-Mount Flange Seat

Drop-in OpenSCAD override:

`hardware/optics/openscad/picamera_hq_cmount.scad`



What changed:

- ▶ `mount_height`: 4.5 → 17.526 mm
- ▶ `sensor_height`: 1 → 0 mm
- ▶ PCB posts → smooth flange seat
- ▶ Cover STL removed (body is its own enclosure)

Geometry:

Flange housing OD (HQ Cam)	32.0 mm
Printed seat ID	32.6 mm
Slip-fit clearance	0.3 mm
Seat depth	4.0 mm
Optical clear bore	26.0 mm

Design Decision — Slip-Fit vs. Printed Thread

Why we did not print a 1"-32 UN female C-mount thread.

Printed thread (rejected)

- ▶ Pitch 0.794 mm at 25.4 mm dia. is below FDM tolerance.
- ▶ Layer adhesion fights the helix; thread strips after a few cycles.
- ▶ Adds modelling complexity and print time without measurable benefit.

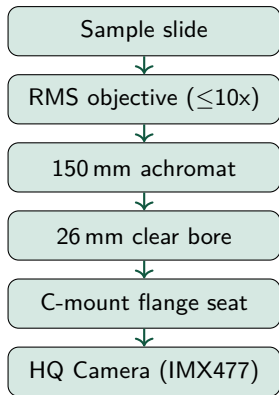
Slip-fit seat (chosen)

- ▶ One smooth bore, robust against print variation.
- ▶ 0.3 mm clearance is tighter than typical FDM thread accuracy.
- ▶ Flange face self-references against printed lip — gravity does the work in inverted geometry.
- ▶ Easy to add a printed retainer if needed.

Both 125 mm and 150 mm STLs regenerated automatically: same camera-side geometry, different tube length above the seat.

Updated Optical Stack

Light path (sample → sensor):



Output STLs (regenerated):

File	Size
optics_hq_150_rms.stl	2.0 MB
optics_hq_125_rms.stl	1.9 MB

Sensor utilisation:

150 mm → effective FN 19.0 mm → **~95 %** of IMX477 active area.

Build: `./build_optics.sh` regenerates both variants and copies the C-mount override automatically.

Roadmap Update

Phase	Deliverable	Status
Hardware base	RPi 5, AI HAT, HQ Cam, OFM server	✓ Operational
Parts procurement	Sangaboard v5, motors, hardware kit	✓ Received
Optics design	Tube lens calc., sensor matching	✓ Verified
Custom optics STL	125 / 150 mm + C-mount seat	✓ Generated
Tube lens delivery	Achromatic doublet, 12.7 mm	✓ Received
3D printing	Body, optics, gears, illumination	○ Next
ISP tuning file	IMX477 in OpenFlexure (black-frame fix)	➔ In progress
Mechanical assembly	Microscope build + camera fit	○ Pending
First light	Sample on stage, image on web UI	○ Pending
Stage integration	Sangaboard wired + SangaboardThing	○ Pending
Time-lapse trial	1 h test run, format → disk	○ Pending
Dataset (500–1000)	Annotated frames, CVAT + SAM2	○ Pending
Model training	YOLOv8n-seg fine-tune	○ Pending
Hailo deployment	Compiled HEF on NPU	○ Pending

What Else This Project Can Deliver

Beyond the immediate experiment, the platform produces reusable artifacts.

For the lab:

- ▶ **Reproducible deploy:** Ansible playbook brings any RPi 5 to "ready" in <10 min.
- ▶ **Benchmarked NPU:** Hailo-8 verified at 91.5 inf/s, 10.9 ms (YOLOv8s Pose).
- ▶ **Dataset pipeline:** CVAT → SAM2 → YOLO → HEF, scriptable end-to-end.
- ▶ **Cost ceiling:** entire scope ~300 EUR; replicable elsewhere.

For the community:

- ▶ **Upstream contribution:** C-mount seat patch is a candidate PR for OpenFlexure (hq_camera branch).
- ▶ **Pi 5 patches:** pisp tuning, pipewire mask, DKMS ordering — already filed.
- ▶ **Open BOM:** every part referenced by datasheet, no proprietary tooling.
- ▶ **HEF zoo:** compiled cell-segmentation model usable on any Hailo-8 device.

Concrete Extensions Worth Discussing

Each is independently scoped — pick what fits the timeline.

Extension	Why	Effort
Incubator-safe enclosure	37 °C/humidity rated; computing stays out	1–2 weeks
Flat-field calibration	Removes vignetting + lens shading from raw frames	2–3 days
Autofocus via Hailo NPU	Sharpness inferred online; closes z-loop	1 week
Time-lapse scheduler	Cron + REST; logs to NAS automatically	2 days
Live overlay on web UI	Inference results streamed back to OFM UI	1 week
Inverted-stage variant	Sample on top, objective below — chip-friendly	STL fork
Multi-channel illumination	Add fluorescence LEDs (calcein for Ca^{2+})	2 weeks + parts

Risks Still on the Table

Risk	Mitigation	Severity
Tube lens BFD off datasheet	tube_lens_ffd is one parameter; reprint optics module	Low
Slip-fit too tight or loose	Adjust seat ID by ± 0.1 mm and rebuild	Low
Camera black-frame (ISP tuning)	rpicam-still works; OpenFlexure venv missing pisp tuning file for IMX477	Open
Objective parfocal $\neq$ 45 mm	35 mm variant available; verify before printing	Low
Bioreactor flow rate undefined	Empirical calibration once chip + pump paired	Medium
Annotation throughput	SAM2-assisted; fall back to coarser classes if too slow	Medium

Immediate Next Steps

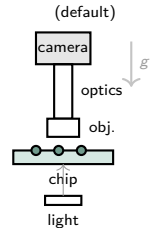
Two-week horizon — hardware and software in parallel:

1. **Print queue:** main body, optics module (150 mm), gears, condenser, illumination, tools (black PLA / PETG, 0.15 mm).
2. **Verify tube lens BFD.** Check 150 mm achromat datasheet against `tube_lens_ffd = 146.9 mm`; adjust and rebuild STL only if different.
3. **ISP tuning file.** `rpicas-still` works; OpenFlexure's `StreamingPiCamera2` produces black frames — `psp` tuning file for IMX477 is missing in the OpenFlexure venv path. Resolve in parallel with printing.
4. **Mechanical assembly.** Insert tube lens, thread objective, slide HQ Camera into C-mount seat.
5. **First light.** Mount on dovetail, route ribbon cable, confirm live stream in OpenFlexure web UI.
6. **Stage integration.** Wire Sangaboard, flash firmware, switch `ofm_config.json` from `DummyStage` → `SangaboardThing`, redeploy via Ansible.
7. **Stage calibration.** Step 28BYJ-48 motors, measure step size with calibration slide.

After step 7 we are at "first useful image" — the gate to dataset collection.

Geometry Constraint — Side Windows Require a Horizontal Axis

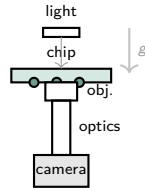
OFM Inverted



Optical axis vertical.

Bottom window needed.

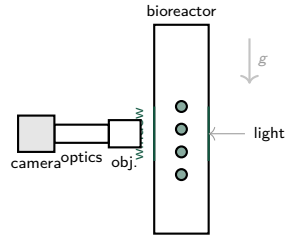
OFM Upright



Optical axis vertical.

Top window needed.

Required: Horizontal



Optical axis horizontal.

Gravity $\perp$ optical axis.

Neither standard OFM orientation works. Both assume gravity $\parallel$ optical axis; the flexure stage relies on this for preload. Rotating 90° puts lateral gravity load on the flexures $\rightarrow$ reduced stiffness and drift.

Options: custom horizontal bracket for OFM · external XYZ stage + camera arm · consult OFM forum.

Potentially Relevant Literature — Garcia Garcia et al., PNAS 2026

Engineered & decellularized human hypertrophic cartilage (D-HyC) as bone graft substitute

Garcia Garcia et al., *PNAS* 2026 doi:10.1073/pnas.2507185123

Core result:

- ▶ MSOD-B hMSCs → hypertrophic cartilage → decellularize
- ▶ >99.7% cell removal; preserves collagen + BMP-2
- ▶ 13/13 critical femoral defect bridging in rats at 12 wks

Relevance for us:

- ▶ BMP-2 + collagen = direct osteogenic cues for hFOB 1.19
- ▶ Validates artificial bone matrix concept in Compartment A
- ▶ SI Fig. S2C/D: hMSC powder supplementation

Practical angle:

D-HyC as scaffold in Compartment A instead of generic sponge.

Caveats:

- ▶ MSOD-B line needed (Pigeot et al. 2021, *Adv. Mat.*); contact Bourguine paul.bourguine@med.lu.se
- ▶ 17.6% GAG loss — note if GAG readouts planned

Discussion: Does D-HyC replace or supplement the sponge matrix in Compartment A?

What I Need From the Group

- ▶ **Decision:** fluorescence channel from the start, or brightfield first?
(Affects illumination kit. Calcein for Ca^{2+} requires 494/517 nm excitation/emission.)
- ▶ **Design input:** the bioreactor's side windows require a horizontal optical axis — neither standard OFM orientation works without modification (see previous slide).
Key question: is the chip position fixed, or can a custom mount bring the OFM objective to the window?
- ▶ **Input:** mineralization reproducibility is the current open question — what are the main variables suspected? (Flow rate? Seeding density? Passage number?)
This drives which parameters to keep fixed in the first imaging protocol.
- ▶ **Resource:** access to a compatible chip + perfusion pump for the first wet test.

Summary

Where we are

Hardware fully procured. Custom optics designed and fabricated as STLs. First obstacle (camera-side interface) identified and resolved.

What's blocking first light

A print run. No *hardware* outstanding. Two open software tasks remain: ISP tuning file for the IMX477 (black-frame issue in OpenFlexure) and Sangaboard stage integration.

What this platform unlocks

Reproducible, low-cost, AI-augmented live-cell imaging. Beyond the hFOB / hMEC experiment, every artifact (STL, ansible role, HEF model) is reusable.

Thank you

Questions, decisions, or new constraints — now is the cheapest moment to factor them in.



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